

Reproduction of karpathy/minbpe: BPE tokenizer training and tests

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Independent reproduction · Benchmark or leaderboard entry · Natural language processing

Abstract

Ran train.py unmodified at the pinned commit: trains the Basic and Regex byte-pair-encoding tokenizers to a 512-token vocabulary on the repository's sample text. The run then loads both saved tokenizers, checks that encoding round-trips the text, measures the compression ratio (bytes per token), and runs the repository's test suite, which compares against tiktoken. CPU only, one machine.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00010v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	BPE training, compression an...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: BPE training, compression and tests

Run	Metric	Final value	Step
1	training_seconds	15.66	
1	merges_logged	512	
1	compression_basic	2.35908	
1	compression_regex	2.12698	
1	tests_passed	21	
1	tests_failed	0	

Run	Inline claim	Value	Phase
1	training_seconds	15.66	
1	compression_basic	2.35908	
1	compression_regex	2.12698	
1	tests_passed	21	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	32.8 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	423654c35fb8b5defd69c3e99b9c16bb8f2fd8fc21e68a973368636889dbae78	281 KB
r1.kvbundle.zip	c99f4be5dff693676a71fd28906bf6b185e89d896fe034894227db3a6aa4a964	17 KB

5 Links and licence

Reproduction of: <https://github.com/karpathy/minbpe/tree/1acefe89412b20245db5a22d2a02001e547dc602>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: tokenization, bpe, minbpe.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (BPE training, compression and tests), kv:2610.00010v1/r1. Data hash 7955540c833ac615d223dba88b38b2a56fdab743ca0f614c9d6a92ec5a5dff80.

<https://kveritas.org/records/2610.00010v1/r1.pdf>